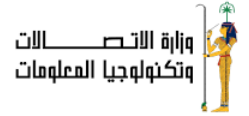




# Project Documentation

**SafeLA Analytics (2020–May 2025)**



## 1.1 Project Proposal

**Project Title:** Crime Analysis & Forecasting Dashboard (2020–May 2025)

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## 1.2 Project Summary

This project analyzes crime incidents reported in Los Angeles from **January 2020 to May 2025**.

We cleaned, transformed, modeled, and visualized the data using multiple tools: Tableau, Power BI, Excel, Python, and forecasting models (CatBoost).

The final output is a set of interactive dashboards that help government agencies, police departments, and analysts identify crime trends, hotspots, victim demographics, weapon usage, and future crime forecasts.

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## 1.3 Objective:

- Analyze crime patterns across time, location, and type.
  - Identify the most affected age groups and genders.
  - Understand weapon usage and arrest patterns.
  - Predict crime levels for the next 2 years.
  - Create interactive dashboards supporting decision-making.
- 

## 1.4 Scope:

Included:

- Data cleaning
- Categorization of 140+ crime types
- Categorization of 300+ weapon types
- Exploratory analysis
- Dashboards in Tableau, Power BI, Excel

- Forecasting using CatBoost

Excluded:

- Real-time prediction
- Detailed case-level qualitative analysis

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## **2. Team Roles & Responsibilities**

| Team Member | Task                                                                    |
|-------------|-------------------------------------------------------------------------|
| Sara        | Team Leader, Tableau dashboards, documentation, PPT, insights           |
| Nariman     | Excel dashboard, Data collection, Power Query cleaning, data dictionary |
| Moataz      | Power BI dashboard, Python cleaning, notebooks, QA validation           |
| Ziad        | Python dashboard, EDA & insights generation, chart prototyping          |
| Alaa        | 2-year forecasting model using CatBoost + Tableau forecast dashboard    |

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## **3. Data Description**

### **3.1 Dataset Source**

Los Angeles Crime Dataset (2020–May 2025)

### **3.2 Data Dictionary**

| Column         | Meaning / What It Represents                                        |
|----------------|---------------------------------------------------------------------|
| Incident_ID    | Unique identifier for each crime report                             |
| Report_Date    | Date crime was officially reported                                  |
| Incident_Date  | Actual date when the crime occurred                                 |
| Incident_Time  | Crime occurrence time                                               |
| Area_Name      | Name of the geographic reporting area                               |
| Crime_Type     | Original detailed crime description (140+ values)                   |
| Victim_Age     | Age of the victim                                                   |
| Victim_Gender  | Gender of the victim                                                |
| Premise_Type   | Location type where crime occurred                                  |
| Case_Status    | Whether arrest happened or not                                      |
| Weapon_Cleaned | Weapon category (after grouping 300+ weapons → 10 clean categories) |
| Latitude       | GPS coordinate                                                      |
| Longitude      | GPS coordinate                                                      |
| Year           | Extracted crime year                                                |
| Month          | Extracted crime month                                               |

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| <b>Day of Week</b>       | Day name (Monday–Sunday)                           |
| <b>Report_Delay_Days</b> | Days between incident & report                     |
| <b>Time_Period</b>       | Categorized time (Morning/Afternoon/Evening/Night) |
| <b>Victim_Age_Group</b>  | Age group category                                 |
| <b>Arrest_Flag</b>       | 1 = Arrested, 0 = Not Detained                     |
| <b>Month_Year</b>        | Monthly time index used for trends                 |

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## **4. Data Cleaning & Transformation**

### **4.1 Cleaning Steps:**

#### **Step 1: Load Data**

Load CSV file and verify it exists.

```
df = pd.read_csv('LADData.csv')
```

---

#### **Step 2: Column Mapping**

Standardize column names for consistency.

##### **Mapping Examples:**

- DR\_NO → dr\_no
  - Date Rptd → date\_rptd
  - DATE OCC → date\_occ
  - Vict Age → vict\_age
-

### Step 3: Column Selection

Keep 19 essential columns, drop 8 unnecessary ones.

#### Essential Columns (19):

| Category       | Columns                                                               |
|----------------|-----------------------------------------------------------------------|
| IDs & Location | dr_no, area, area_name, rpt_dist_no, location, cross_street, lat, lon |
| Date & Time    | date_rptd, date_occ, time_occ                                         |
| Crime Info     | crm_cd, crm_cd_desc, status, status_desc                              |
| Victim Info    | vict_age, vict_sex, vict_descent                                      |
| Details        | premis_desc, weapon_desc                                              |

### Step 4: Missing Values Analysis

Document missing data patterns without deleting rows.

#### Common Patterns:

- weapon\_desc: High missing (not all crimes involve weapons)
- cross\_street: Moderate missing (not all locations at intersections)
- vict\_age/sex/descent: Some missing (unknown victim info)

```
=====
4. MISSING VALUES ANALYSIS
=====
```

|              | Missing_Count | Percentage |
|--------------|---------------|------------|
| crm_cd_4     | 1004927       | 99.993632  |
| crm_cd_3     | 1002677       | 99.769749  |
| crm_cd_2     | 935831        | 93.118346  |
| cross_street | 850755        | 84.652997  |
| weapon_cd    | 677744        | 67.437818  |
| weapon_desc  | 677744        | 67.437818  |
| mocodes      | 151619        | 15.086603  |
| vict_descent | 144656        | 14.393761  |
| vict_sex     | 144644        | 14.392567  |
| premis_desc  | 588           | 0.058508   |
| premis_cd    | 16            | 0.001592   |
| crm_cd_1     | 11            | 0.001095   |
| status       | 1             | 0.000100   |

## **Step 5: Date/Time Cleaning**

### **5.1 Date Fields**

**Fields:** date\_rptd, date\_occ

#### **Invalid Criteria:**

- X Future dates (> today)
- X Before 1900
- X Null/missin

### **5.2 Time Field**

**Field:** time\_occ (Military format: HHMM)

#### **Validation:**

- Must be 4 digits
- Hour: 00-23
- Minute: 00-59

### **5.3 Combined DateTime**

Create datetime\_occ by combining valid date + time for time-series analysis.

---

## Step 6: Victim Age Cleaning

**Valid Range:** 0 to 120 years

**Invalid if:**

- Age < 0 (negative)
- Age > 120 (unrealistic)
- Null/missing

```
=====
6. CLEANING VICTIM AGE
=====
Age statistics:
  Min: -4
  Max: 120
  Mean: 28.92
  Median: 30.0
Invalid ages (< 0 or > 120 or null): 137

Top problematic age values:
vict_age
-1      100
-2       28
-3        6
-4         3
Name: count, dtype: int64
```

---

## Step 7: Categorical Fields Standardization

### 7.1 Victim Sex

**Valid Codes:** F, M, X, NAN, NONE

**Process:**

1. Convert to uppercase
2. Trim whitespace
3. Validate against allowed values

```
=====
7. CLEANING CATEGORICAL FIELDS
=====
Vict Sex - Invalid values: 115
  Unique invalid values: ['H' '-']
  Value counts:
vict_sex
M      403879
F      358580
NAN    144644
X       97773
H        114
-         1
Name: count, dtype: int64
```

---

### 7.2 Victim Descent

**Valid Codes:** Single letters A-Z plus special codes

**Common Codes:**

- H = Hispanic/Latin/Mexican
- W = White
- B = Black
- A = Other Asian
- X = Unknown

- *[Full list: A,B,C,D,F,G,H,I,J,K,L,O,P,S,U,V,W,X,Z]*

```
Vict Descent - Invalid values: 0
```

```
Value counts:
```

```
vict_descent
```

```
H      296404
```

```
W      201442
```

```
NAN    144656
```

```
B      135816
```

```
X      106685
```

```
O       78005
```

```
A       21340
```

```
K        5990
```

```
F        4838
```

```
C        4631
```

```
J        1586
```

```
V        1195
```

```
I        1015
```

```
Z         577
```

```
P         288
```

```
U         221
```

```
D          91
```

```
L          77
```

```
G          74
```

```
S          58
```

```
-           2
```

```
Name: count, dtype: int64
```



## Step 8: Geographic Data Cleaning

### LA County Boundaries:

- Latitude: 33.7 to 34.35
- Longitude: -118.7 to -118.1

```
=====
8. CLEANING GEOGRAPHIC DATA
=====
Latitude range: 0.0000 to 34.3343
Longitude range: -118.6676 to 0.0000
Invalid coordinates: 2240
  Zero coordinates (0,0): 2240
  Out of LA bounds: 0
```

---

## Step 9: Duplicate Detection

Check for duplicate DR\_NO values (should be unique).

**Action:** Report count but don't delete (may be legitimate case updates)

---

## Step 10: Crime Code Validation

Validate that crm\_cd matches crm\_cd\_1.

### Report:

- Count of mismatches
- Top 10 most common crime codes
- Crime type distribution

```
=====
10. VALIDATING CRIME CODES
=====
Crime code mismatches (crm_cd vs crm_cd_1): 1004991

Top 10 Crime Codes:
crm_cd
510    115190
624    74839
330    63517
354    62537
740    61092
310    57871
440    53717
230    53525
626    46712
420    41314
Name: count, dtype: int64
```

---

## Step 11: Text Field Cleaning

**Fields Cleaned:** location, cross\_street, area\_name, premis\_desc, weapon\_desc, status\_desc, crm\_cd\_desc

### Operations:

1. Trim leading/trailing spaces
  2. Replace 'nan' strings with actual NaN
  3. Replace 'None' strings with NaN
-

## Step 12: Area Validation

**Valid Range:** LAPD areas 1-21

**Invalid if:**

- Area < 1
- Area > 21
- Null/missing

**The 21 Areas:** Central, Rampart, Southwest, Hollenbeck, Harbor, Hollywood, Wilshire, West LA, Van Nuys, West Valley, Northeast, 77th Street, Newton, Pacific, N Hollywood, Foothill, Devonshire, Southeast, Mission, Olympic, Topanga

### 12. VALIDATING AREA CODES

Invalid area codes (should be 1-21): 0

Area distribution:

area

|    |       |
|----|-------|
| 1  | 69670 |
| 2  | 46825 |
| 3  | 57441 |
| 4  | 37085 |
| 5  | 41394 |
| 6  | 52429 |
| 7  | 48239 |
| 8  | 45729 |
| 9  | 42883 |
| 10 | 42156 |
| 11 | 42963 |
| 12 | 61758 |
| 13 | 49177 |
| 14 | 59514 |
| 15 | 51107 |
| 16 | 33133 |
| 17 | 41756 |
| 18 | 49936 |
| 19 | 40351 |
| 20 | 50071 |
| 21 | 41374 |

Name: count, dtype: int64

## Saved Files:

```
=====
15. SAVING CLEANED DATA
  Columns: 20 (essential columns only)
  Purpose: Production-ready for dashboards and analysis

✓ Saved: LADData_issues.csv
  Records: 2,492 (0.2% of total)
  Columns: 27 (20 data + 7 flags)
  Purpose: Manual review and data quality investigation

=====
FINAL SUMMARY
=====
Original records: 1,004,991
High quality records: 1,002,499 (99.8%)
Records with issues: 2,492 (0.2%)

Output files created (3 files):
  1. LADData_with_flags.csv
    - All records with quality flags
    - 27 columns
    - Use for: Quality control

  2. LADData_high_quality.csv ★ USE THIS
    - Clean records only (1,002,499 records)
    - 20 columns
    - Use for: Dashboards, visualizations, analysis

  3. LADData_issues.csv
    - Problematic records only (2,492 records)
    - 27 columns (includes flags)
    - Use for: Data quality review

✓ Data cleaning pipeline complete!
=====
```

---

## Step 13: Handling Invalid Age Values (0)

### Problem

Some rows contained an **Age = 0**, which is not a valid age.

### Method

1. Used Power Query to calculate the **median age**, which was **30**.
2. Replaced all age values equal to **0** with the median (30).

### Reasoning

- Median is more resistant to outliers than the mean.
- Ensures age distribution stays realistic.

## Step 14: Fixing Missing Values in the Descent Column

### Problem

The *Descent* column contained missing values (NaN).

### Method

1. Identified the **mode** (most frequent value) using a **Pivot Table** in Excel.
2. Replaced all NaN values in *Descent* with that mode.

### Reasoning

- Replacing categorical nulls with the mode preserves the distribution of categories.
  - Prevents category loss and reduces bias.
- 

## Step 15: Handling Null Values in the Sex Column

### Problem

The *Sex* column contained null values.

### Method

1. Used an Excel Pivot Table to count the distribution of Male vs. Female.
2. Calculated the most common value (mode), which was **Male**.
3. Replaced all null values in the *Sex* column with “**Male**”.

### Reasoning

- For categorical binary variables, mode is the most appropriate imputation.
  - Prevents imbalance and keeps dataset consistent.
- 

## Step 16: Validation

All replacement steps were checked using:

- Preview of results in Power Query
  - Pivot Tables in Excel
  - Screenshots (recorded for documentation)
-

## Step 17: Time Column Cleaning

The `time_occ` column contained inconsistent and invalid time formats such as missing digits, mixed characters, and impossible time values. To standardize this column, I wrote a Python function that automatically cleans and converts all entries into a uniform **HH:MM** format.

### Steps Performed

- 1. Extract Only Numeric Characters**
    - a. Some values included symbols or text.
    - b. The function keeps only digits and removes everything else.
  - 2. Fix Incomplete Time Values**
    - a. Short values like 30 or 845 were padded with zeros to a 4-digit format.
      - i. Example: 30 → 0030, 845 → 0845.
  - 3. Split Into Hours and Minutes**
    - a. The first two digits become the hour, the last two the minutes.
  - 4. Validate Time**
    - a. Hours greater than 23 or minutes greater than 59 were marked as invalid and replaced with null.
  - 5. Convert to Standard Time Format (HH:MM)**
    - a. All valid values were formatted correctly (e.g., 0845 → 08:45).
  - 6. Convert Column to Datetime Type**
    - a. Ensures the column is sortable and ready for time-based analysis.
  - 7. Save the Cleaned Dataset**
    - a. The cleaned dataset was exported to a new CSV file.
- 

## 4.2 Validation Rules

### Summary Table

| Field       | Valid Criteria      | Invalid Examples             | Flag                 |
|-------------|---------------------|------------------------------|----------------------|
| Dates       | 1900 < date < today | Future dates, pre-1900, null | date_flag            |
| Time        | 0000-2359 format    | 2400, 1265, abcd             | time_valid           |
| Age         | 0-120 years         | -1, 150, null                | age_flag             |
| Sex         | F, M, X, NAN, NONE  | male, f, invalid             | sex_flag             |
| Descent     | A-Z single letters  | AA, invalid, null            | descent_flag         |
| Coordinates | LA County bounds    | (0,0), out of range, null    | coord_flag           |
| Area        | 1-21                | 0, 25, null                  | <i>Reported only</i> |

## 4.3 Output Files

### **File 1: LAData\_with\_flags.csv**

**Purpose:** Quality control and debugging

**Contents:**

- All records (100%)
  - 26 columns: 19 data + 7 quality flags
  - Both clean and problematic records
- 

### **File 2: LAData\_high\_quality.csv**

**Purpose:** Production-ready for analysis and visualization

**Contents:**

- Clean records only (~90%)
  - 19 data columns (NO quality flags)
  - All validation checks passed
- 

### **File 3: LAData\_issues.csv**

**Purpose:** Manual review and quality investigation

**Contents:**

- Problematic records only (~10%)
  - 26 columns: 19 data + 7 quality flags
  - Records that failed validation
-

## 4.4 Process Summary

### Final Statistics

- **Input:** 27+ columns
  - **Output:** 19 data columns + 7 quality flags
  - **Clean Records:** ~90%
  - **Flagged Records:** ~10%
  - **Processing:** 14 steps, fully automated
- 

## 4.5 Crime Type Grouping

We grouped **140+ messy crime descriptions** into **12 clean categories**:

| New Category          | Contains (Examples)                             |
|-----------------------|-------------------------------------------------|
| Assault / Violence    | ASSAULT, BATTERY, THREATS, STALKING, KIDNAPPING |
| Theft / Larceny       | THEFT, BURGLARY, SHOPLIFTING, PURSE SNATCHING   |
| Robbery               | ROBBERY, ATTEMPTED ROBBERY                      |
| Fraud / Forgery       | FRAUD, FORGERY, EMBEZZLEMENT, COUNTERFEIT       |
| Drugs / Narcotics     | DRUGS, NARCOTICS                                |
| Weapons / Firearms    | FIREARM, GUN, SHOTS FIRED, BOMB                 |
| Sex Crimes            | RAPE, SODOMY, LEWD ACTS, PORNOGRAPHY            |
| Child / Family Crimes | CHILD ABUSE, NEGLECT                            |
| Property Damage       | VANDALISM, ARSON                                |
| Vehicle Crimes        | VEHICLE THEFT, BIKE STOLEN                      |
| Public Disorder       | RIOT, DISTURBANCE                               |
| Other / Unknown       | The rest                                        |

---



## 4.6 Weapon Type Grouping

**300+ weapons** grouped into **10 categories**:

| Category               | Includes (Examples)              |
|------------------------|----------------------------------|
| <b>Firearm</b>         | Pistol, Rifle, Shotgun, Revolver |
| <b>Sharp Object</b>    | Knife, Blade, Scissors           |
| <b>Heavy Object</b>    | Hammer, Brick, Pipe              |
| <b>Chemical</b>        | Acid, Caustic substances         |
| <b>Explosive</b>       | Bomb, Device                     |
| <b>Strong-Arm</b>      | Hands, Kicks, Fists              |
| <b>Toy / Simulated</b> | Fake gun                         |
| <b>Vehicle</b>         | Car used as weapon               |
| <b>Animal</b>          | Dog                              |
| <b>Unknown</b>         | Not reported                     |

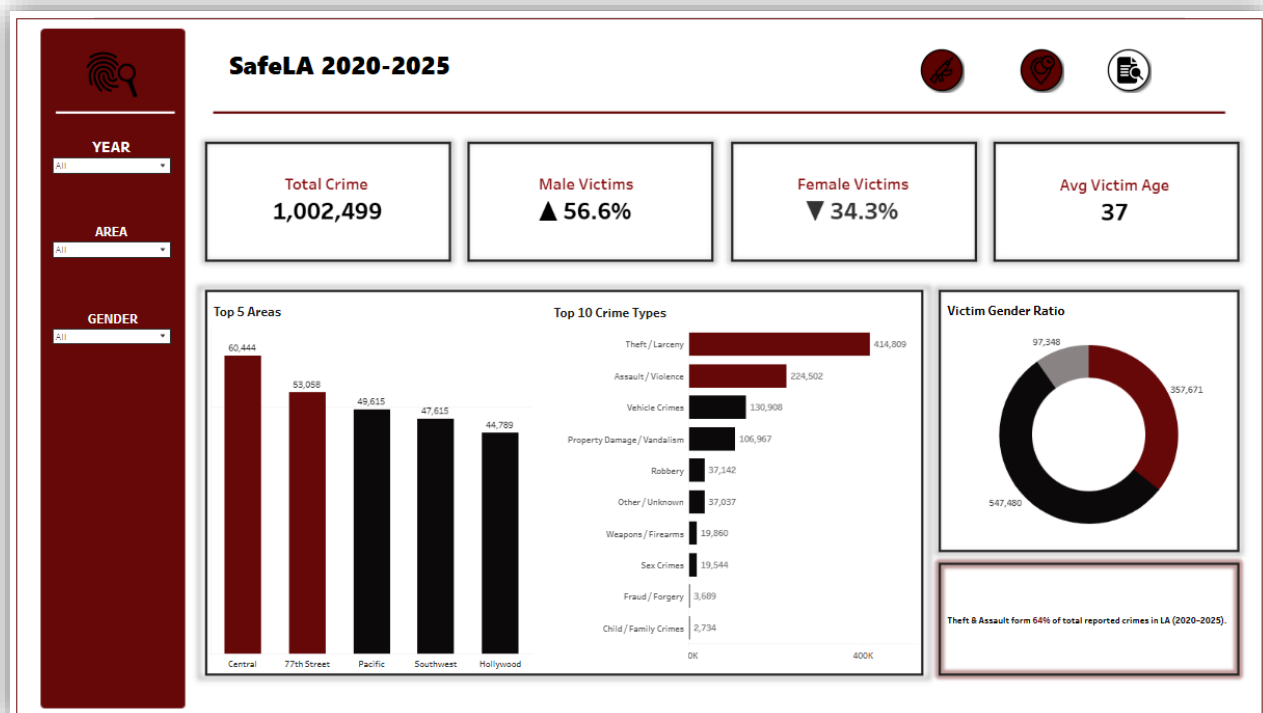
---

## 5. Dashboards Explanation

### 5.1 Tableau Story (Sarah)

#### ❖ Page 1 — Crime Overview

**Objective:** Show the big picture.

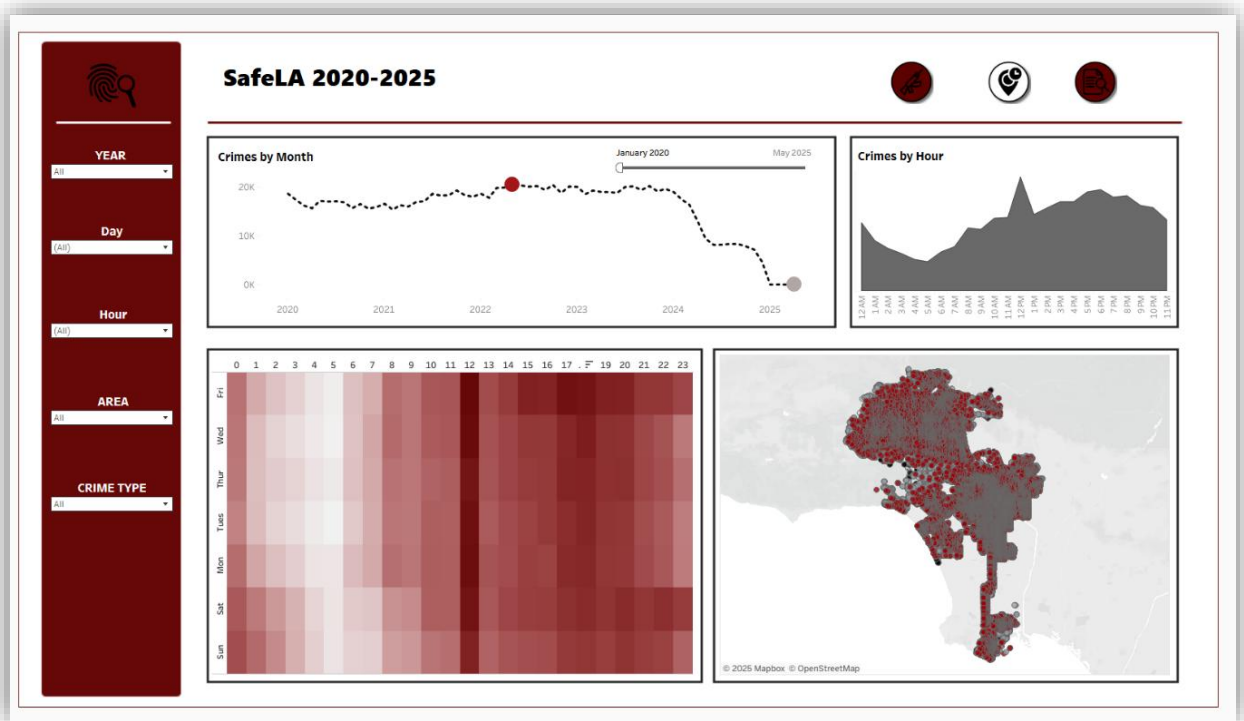


#### Charts:

- **KPI Cards** → Total Crimes, Avg Victim Age, %Male, %Arrested
- **Bar Chart** → Top 10 Crime Types
- **Bar Chart** → Top 10 Areas
- **Pie Chart** → Victim Gender
- **Insight Box** → “Assault + Theft represent 34% of total crimes (2020–May 2025).”

## ❖ Page 2 — Time & Location

**Objective:** When & where crimes happen.

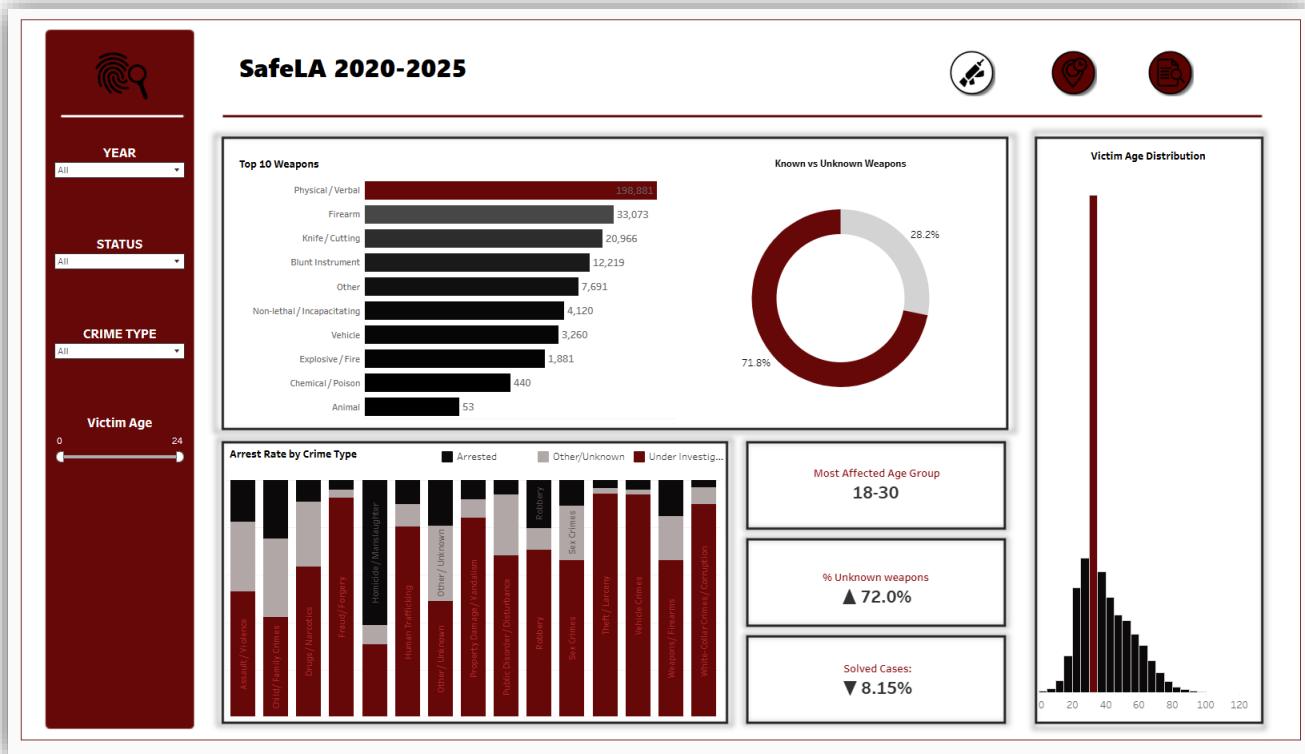


### Charts:

- **Line Chart** → Monthly trends (crimes by month)
- **Histogram** → Crimes by hour
- **Heatmap** → Crimes by hour × weekday
- **Map** → Hotspot areas.

## ❖ Page 3 — Victims, Weapons & Arrests

**Objective:** Who is affected & what happens after the crime.

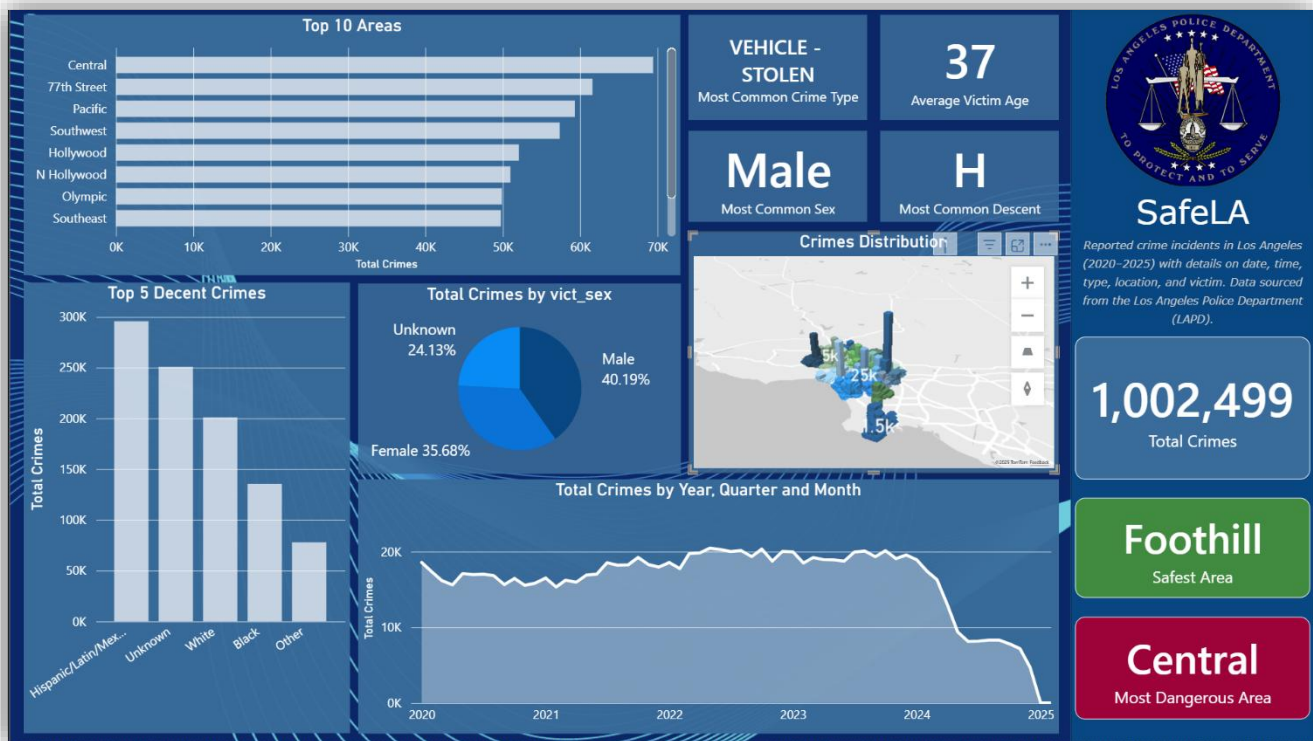


### Charts:

- **KPI Cards** → Most affected age group, %Unknown weapons, %Solved cases
- **Bar Chart** → Top 10 weapons
- **Pie Chart** → Known vs Unknown weapons
- **Histogram** → Victim age distribution
- **Stacked bar** → Arrests by crime type.

Link to dashboard: [https://drive.google.com/drive/folders/1y-qiS7NaaSibKBHs-gVZ5IP2wrgpdFg1?usp=drive\\_link](https://drive.google.com/drive/folders/1y-qiS7NaaSibKBHs-gVZ5IP2wrgpdFg1?usp=drive_link)

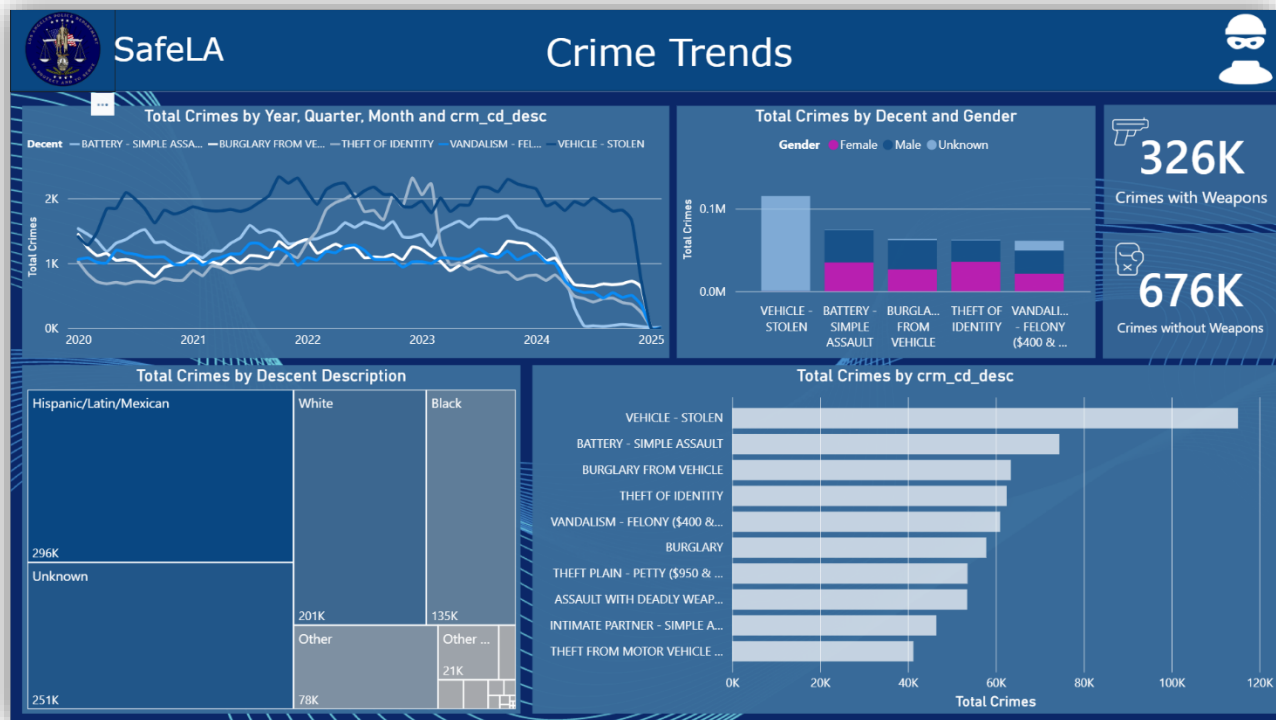
## 5.2 Power BI Dashboard (Moataz)



### A. Overview Page

This page provides a general summary of crimes across the city.

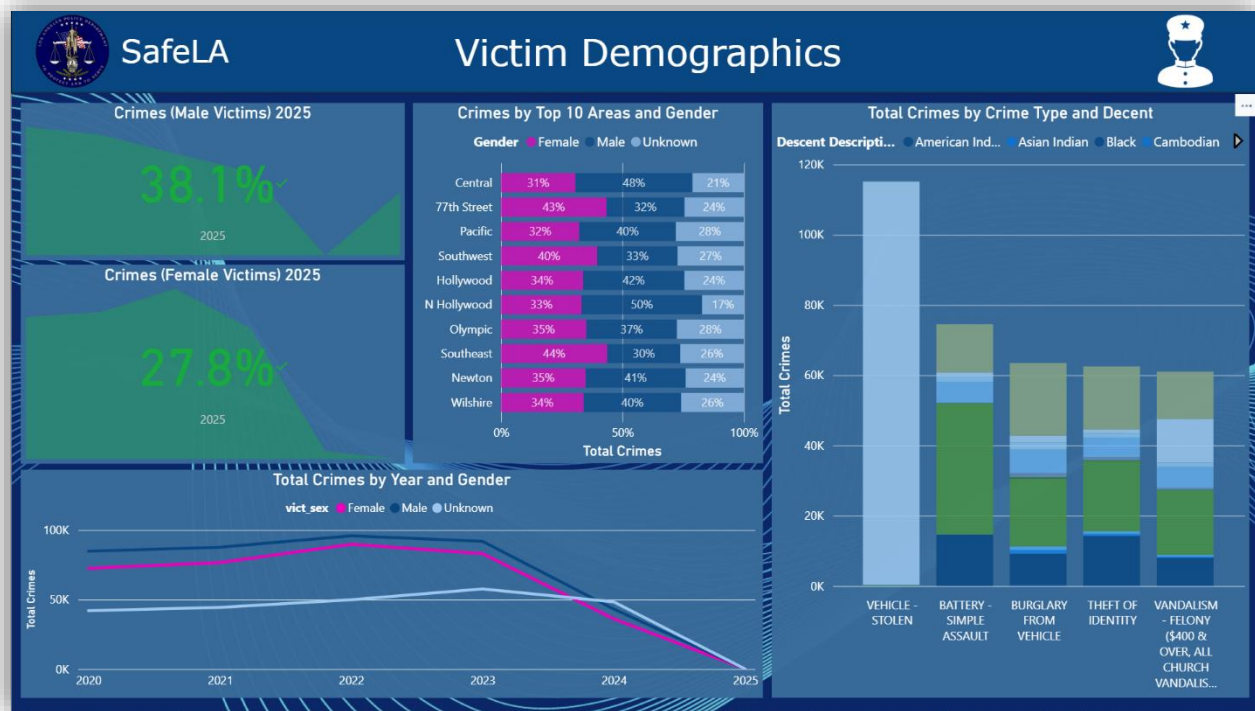
- **Top 10 Areas:** Ranks areas by total crime volume.
- **Most Common Crime Type:** Highlights the highest-occurring crime (e.g., Vehicle – Stolen).
- **Victim Insights:** Shows average victim age, most common sex, and most common descent.
- **Geographical Crime Distribution Map:** Interactive map displaying crime density.
- **Total Crimes Over Time:** Line chart showing yearly and monthly crime progression.
- **Safest & Most Dangerous Areas:** Automatically identifies the lowest and highest crime regions.



## B. Crime Trends Page

Focuses on how crimes evolve over time and differences across groups.

- **Crimes by Type Over Time:** Line chart comparing trends in major crime categories.
- **Crimes by Descent & Gender:** Stacked bar chart summarizing victim characteristics.
- **Weapons Involvement Summary:** Total crimes involving weapons vs. without.
- **Descent Description Tree Map:** Shows the distribution of crimes by descent.
- **Top Crime Types:** Bar chart ranking the most frequent crime categories.



## C. Victim Demographics Page

Provides demographic patterns among victims.

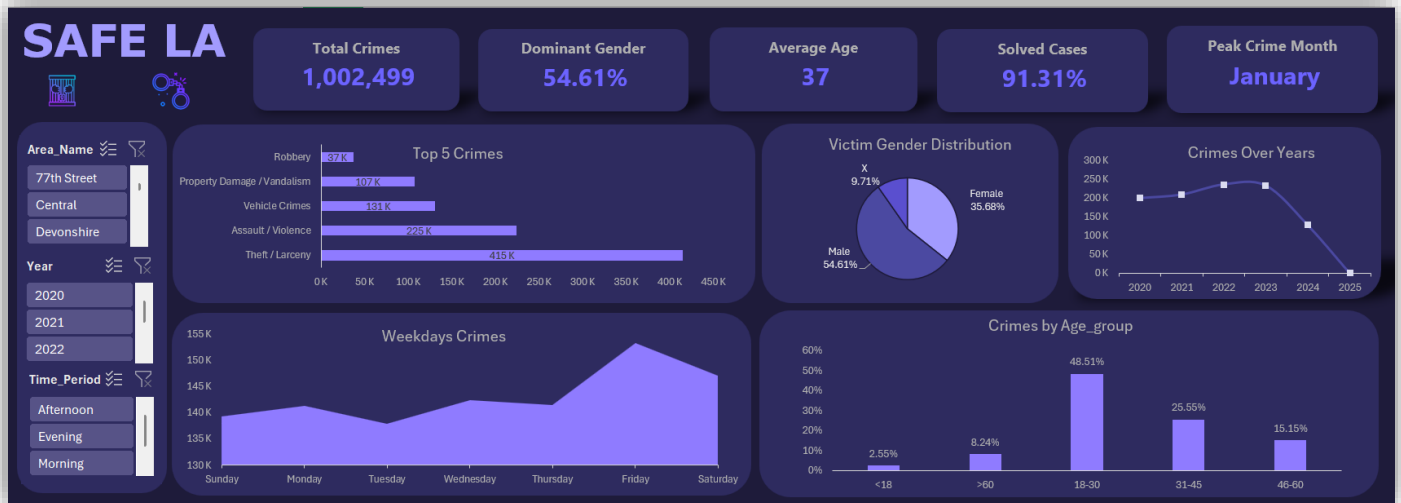
- **Male and Female Crimes (2025):** Summary cards showing percentages by gender.
- **Crimes by Area and Gender:** Horizontal stacked bars comparing gender distribution in the top crime areas.
- **Crimes by Crime Type & Descent:** Stacked bars showing demographic variations across different crime types.
- **Crimes by Year and Gender:** Line chart illustrating changes in victim gender distribution across the years.

Link to dashboard:

[https://drive.google.com/drive/folders/1leXSpX6J672uMhnb6nOkxgA-mj0p-70p?usp=drive\\_link](https://drive.google.com/drive/folders/1leXSpX6J672uMhnb6nOkxgA-mj0p-70p?usp=drive_link)

## 5.3 Excel Dashboard (Nariman)

- Pivot tables
- Crime KPIs
- Year-wise summaries
- Gender distribution
- Area-level stats



### A. Key Performance Indicators

- Total Crimes: 1,002,499
- Dominant Gender: Male (54.61%)
- Average Age: 37
- Solved Cases: 91.31%
- Peak Crime Month: January

### B. Filters

- Area Name
- Year (2020–2022)
- Time Period (Morning, Afternoon, Evening)

### C. Visual Insights

- **Top 5 Crimes:** Vehicle theft, simple assault, vehicle burglary, identity theft, felony vandalism
- **Gender Breakdown:** Male (54.61%), Female (35.68%), Other (9.71%)



- **Yearly Trend:** Crime increases until 2023–2024, then drops sharply in 2025, and that's because 2025 data is not completed yet.
- **Weekday Trend:** Friday has the highest crime levels
- **Age Groups:** Most crimes involve individuals aged 18–30

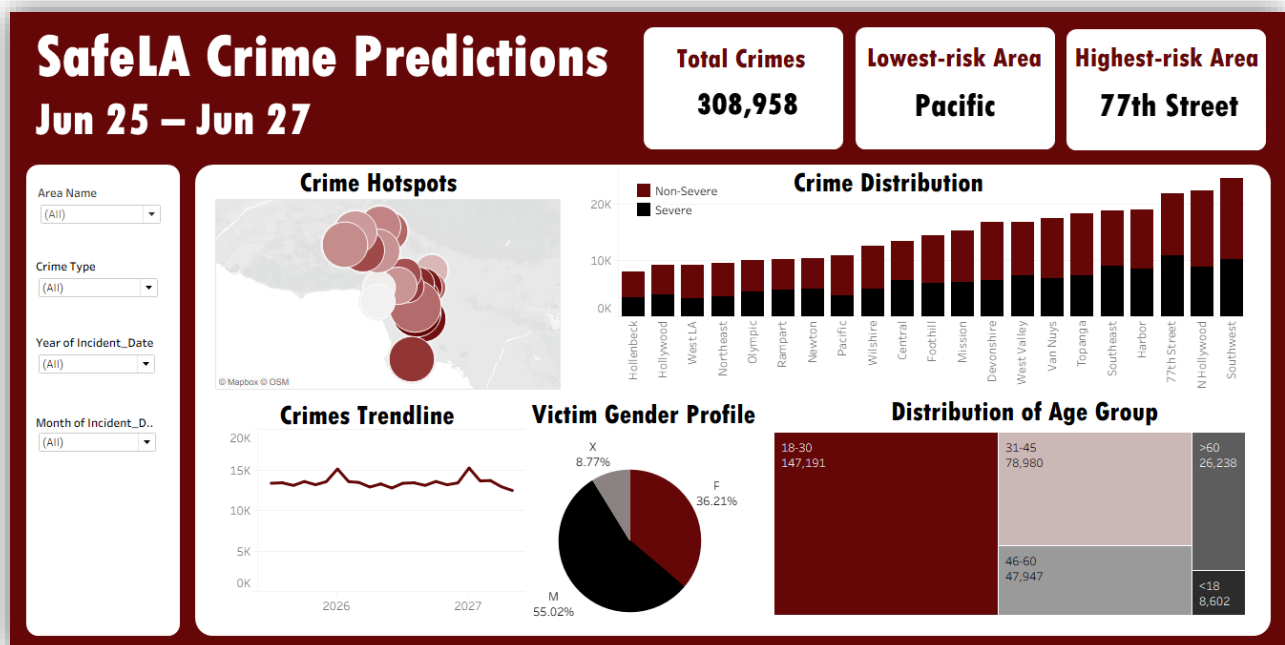
## D. Key Insights

- Vehicle-related crime is the most prevalent category.
- Crime is most common among adults aged 18–30.
- Fridays consistently have the highest crime activity.
- January is the month with the highest number of crimes.

Link to dashboard:

[https://drive.google.com/drive/folders/1k2eO6GKv46lG7hGOEo2DGRX8kOjmrCS?usp=drive\\_link](https://drive.google.com/drive/folders/1k2eO6GKv46lG7hGOEo2DGRX8kOjmrCS?usp=drive_link)

## 5.4 Forecasting Model (Alaa)



- **Preprocessing**
    - Grouped incidents by Area\_Name × Crime\_Type × Incident\_Date.
    - Engineered time-series features: Year, Month, ISO Week, Day.
    - Added lag features (lag\_1, lag\_2) for each area/crime-type series.
  - **Crime Count Model**
    - Trained a CatBoostRegressor to forecast daily crime counts for the next two years.
    - Performed hyperparameter tuning to improve accuracy and avoid overfitting, achieving ~80% accuracy.
  - **Correlation**
    - Encoded categorical variables and calculated correlations to find strongly related features.
    - Displayed correlations using a heatmap for clearer interpretation.
  - **Columns Predictions**
    - Based on feature correlations, set a sequential prediction order:  
Weapon\_Cleaned → Victim\_Age\_Group → Victim\_Gender → Premise\_Type.
    - Built a pipeline where each CatBoostClassifier uses both original features and predictions from earlier targets.
  - **Final Output**
    - Generated a CSV containing:  
Incident\_Date, Area\_Name, Crime\_Type, Weapon\_Cleaned,  
Victim\_Age\_Group, Victim\_Gender, Premise\_Type.
- 

## Charts:

- KPIs: Total Predicted Crimes, Lowest-risk Area, Highest-risk Area
- Crime Hotspots symbol map
- Crimes Trendline
- Crime Distribution across areas
- Victim Gender Profile
- Distribution of Age Group

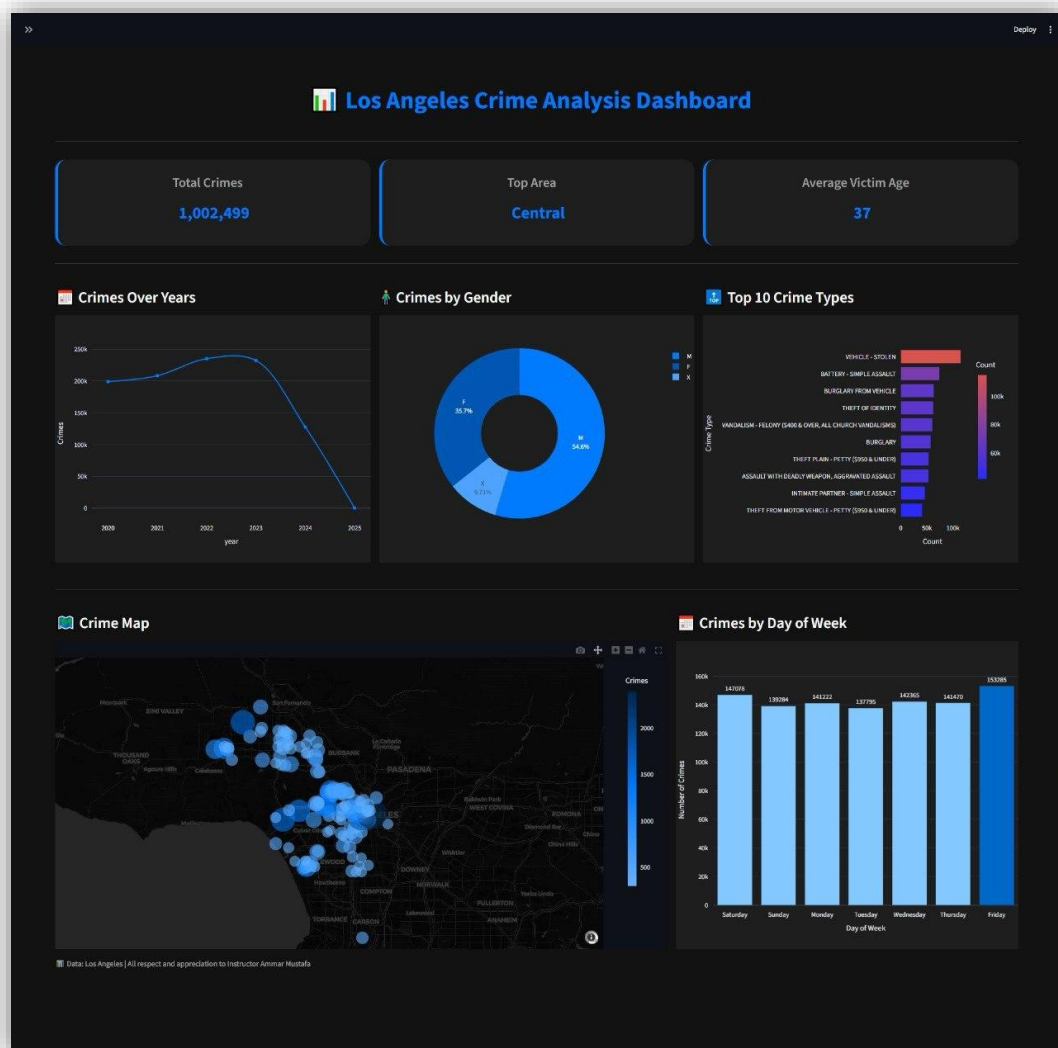
## Insights:

- **Total Crimes:** ~309K
- **Highest-risk Area:** 77th Street; **Lowest-risk Area:** Pacific.
- **Hotspots:** Central and southern LA
- **Trend:** Stable with minor spikes.
- **Victims:** Mostly males (55%) and aged 18–30 (~147k).

**Link to dashboard:**

[https://drive.google.com/drive/folders/1ceo8hxEmniOBS5OfqOyhlg2M7wK6CsfS?usp=drive\\_link](https://drive.google.com/drive/folders/1ceo8hxEmniOBS5OfqOyhlg2M7wK6CsfS?usp=drive_link)

## 5.5 Python Dashboard (Ziad)



## **Insights:**

- Visualizations using Streamlit
- Crime trends
- Age distribution
- Correlation heatmap
- Weapon relationships

## **Charts:**

- KPI Cards: Total Crimes, Top Area, Avg Victim Age
- Line Chart: Crimes Over Years
- Pie Chart: Victim Gender
- Bar Chart: Top 10 Crime Types
- Bar Chart: Crimes by Day of Week
- Insights: Central records the highest crime concentration, with a clear peak in incidents on Fridays.

## **Link to dashboard:**

[https://drive.google.com/drive/folders/1Ot1nuwCV51PCxif0NOiq6PWefrGkMQ9l?usp=drive link](https://drive.google.com/drive/folders/1Ot1nuwCV51PCxif0NOiq6PWefrGkMQ9l?usp=drive_link)

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